

Genome and read alignments

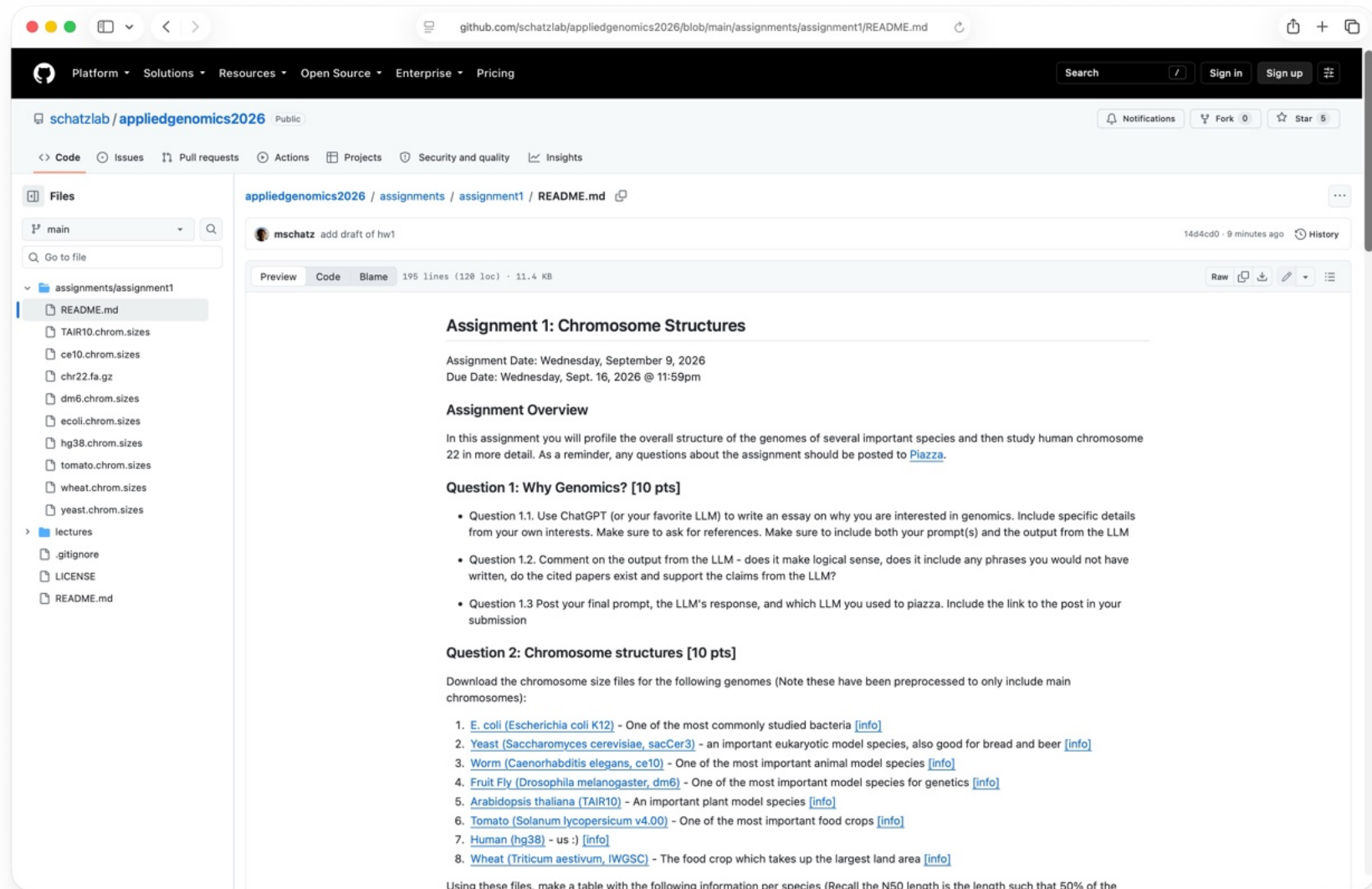
Michael Schatz

Sept 21, 2026

Lecture 6: Applied Comparative Genomics



Assignment I



github.com/schatzlab/appliedgenomics2026/blob/main/assignments/assignment1/README.md

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Files

main

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assignments/assignment1

- README.md
- TAIR10.chrom.sizes
- ce10.chrom.sizes
- chr22.fa.gz
- dm6.chrom.sizes
- ecoli.chrom.sizes
- hg38.chrom.sizes
- tomato.chrom.sizes
- wheat.chrom.sizes
- yeast.chrom.sizes

lectures

- .gitignore
- LICENSE
- README.md

appliedgenomics2026 / assignments / assignment1 / README.md

mschatz add draft of hw1 14d4cd0 · 9 minutes ago History

Preview Code Blame 195 lines (120 loc) · 11.4 KB

Raw Download Edit

Assignment 1: Chromosome Structures

Assignment Date: Wednesday, September 9, 2026
Due Date: Wednesday, Sept. 16, 2026 @ 11:59pm

Assignment Overview

In this assignment you will profile the overall structure of the genomes of several important species and then study human chromosome 22 in more detail. As a reminder, any questions about the assignment should be posted to [Piazza](#).

Question 1: Why Genomics? [10 pts]

- Question 1.1. Use ChatGPT (or your favorite LLM) to write an essay on why you are interested in genomics. Include specific details from your own interests. Make sure to ask for references. Make sure to include both your prompt(s) and the output from the LLM
- Question 1.2. Comment on the output from the LLM - does it make logical sense, does it include any phrases you would not have written, do the cited papers exist and support the claims from the LLM?
- Question 1.3 Post your final prompt, the LLM's response, and which LLM you used to piazza. Include the link to the post in your submission

Question 2: Chromosome structures [10 pts]

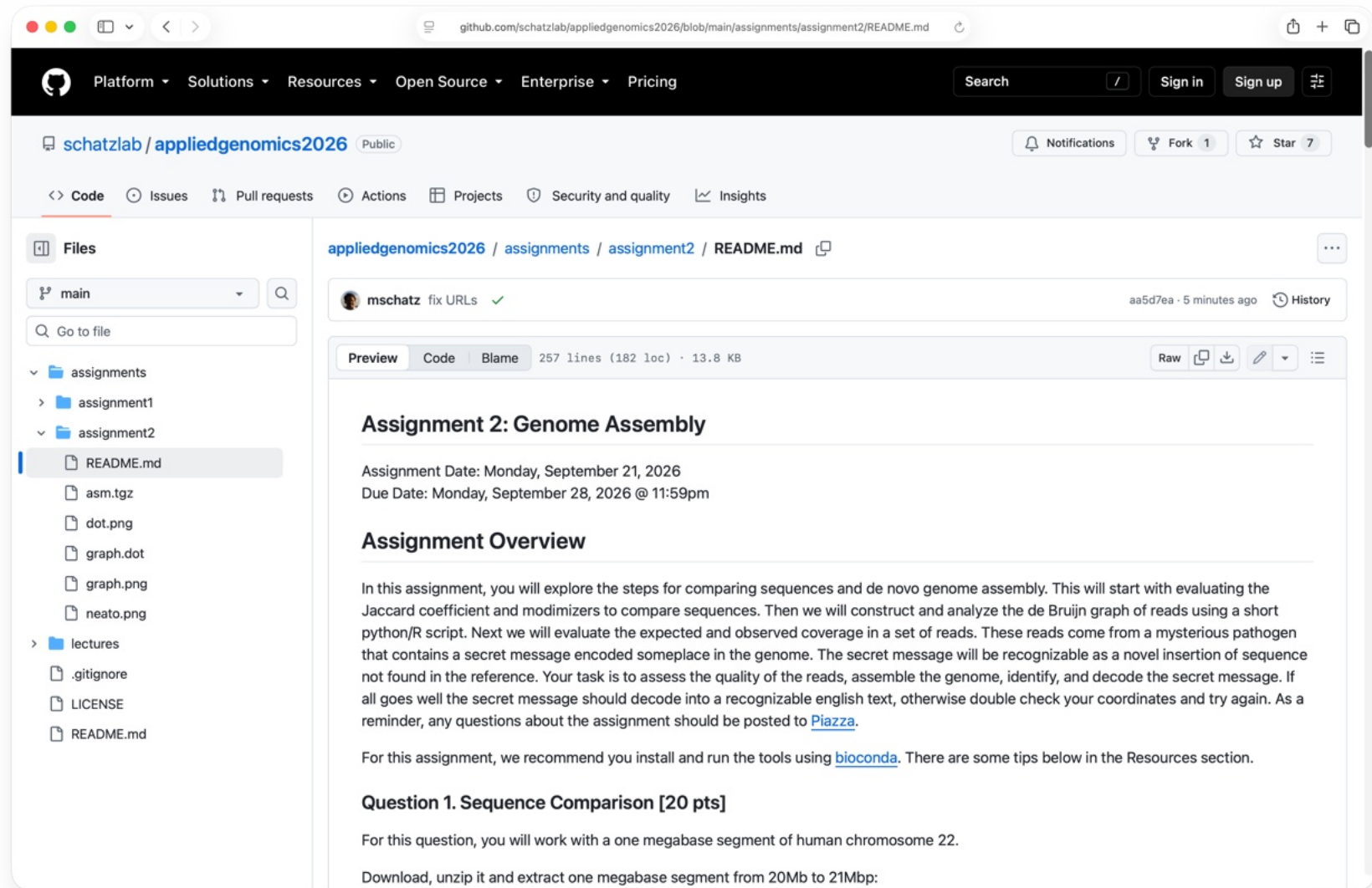
Download the chromosome size files for the following genomes (Note these have been preprocessed to only include main chromosomes):

- [E. coli \(Escherichia coli K12\)](#) - One of the most commonly studied bacteria [\[info\]](#)
- [Yeast \(Saccharomyces cerevisiae, sacCer3\)](#) - an important eukaryotic model species, also good for bread and beer [\[info\]](#)
- [Worm \(Caenorhabditis elegans, ce10\)](#) - One of the most important animal model species [\[info\]](#)
- [Fruit Fly \(Drosophila melanogaster, dm6\)](#) - One of the most important model species for genetics [\[info\]](#)
- [Arabidopsis thaliana \(TAIR10\)](#) - An important plant model species [\[info\]](#)
- [Tomato \(Solanum lycopersicum v4.00\)](#) - One of the most important food crops [\[info\]](#)
- [Human \(hg38\)](#) - us :) [\[info\]](#)
- [Wheat \(Triticum aestivum, IWGSC\)](#) - The food crop which takes up the largest land area [\[info\]](#)

Using these files, make a table with the following information per species (Recall the N50 length is the length such that 50% of the

<https://github.com/schatzlab/appliedgenomics2026/tree/main/assignments/assignment1>
Due end of day on Wednesday Sept 16 (right before midnight)

Assignment 2



The screenshot shows a GitHub repository page for 'appliedgenomics2026'. The left sidebar displays a file tree with folders 'assignments', 'lectures', and files like 'README.md', 'asm.tgz', 'dot.png', 'graph.dot', 'graph.png', 'neato.png', '.gitignore', 'LICENSE', and 'README.md'. The main content area shows the 'assignment2/README.md' file, which includes the title 'Assignment 2: Genome Assembly', the assignment date (Monday, September 21, 2026), the due date (Monday, September 28, 2026 @ 11:59pm), and an 'Assignment Overview' section. The overview describes the task of comparing sequences and de novo genome assembly, mentioning the Jaccard coefficient, modimizers, and a secret message encoded in the genome. It also provides a recommendation to install and run tools using 'bioconda'.

Assignment 2: Genome Assembly

Assignment Date: Monday, September 21, 2026
Due Date: Monday, September 28, 2026 @ 11:59pm

Assignment Overview

In this assignment, you will explore the steps for comparing sequences and de novo genome assembly. This will start with evaluating the Jaccard coefficient and modimizers to compare sequences. Then we will construct and analyze the de Bruijn graph of reads using a short python/R script. Next we will evaluate the expected and observed coverage in a set of reads. These reads come from a mysterious pathogen that contains a secret message encoded someplace in the genome. The secret message will be recognizable as a novel insertion of sequence not found in the reference. Your task is to assess the quality of the reads, assemble the genome, identify, and decode the secret message. If all goes well the secret message should decode into a recognizable english text, otherwise double check your coordinates and try again. As a reminder, any questions about the assignment should be posted to [Piazza](#).

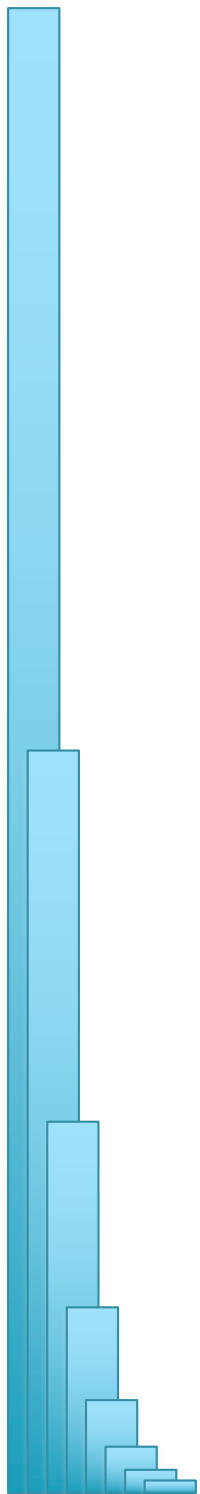
For this assignment, we recommend you install and run the tools using [bioconda](#). There are some tips below in the Resources section.

Question 1. Sequence Comparison [20 pts]

For this question, you will work with a one megabase segment of human chromosome 22.

Download, unzip it and extract one megabase segment from 20Mb to 21Mbp:

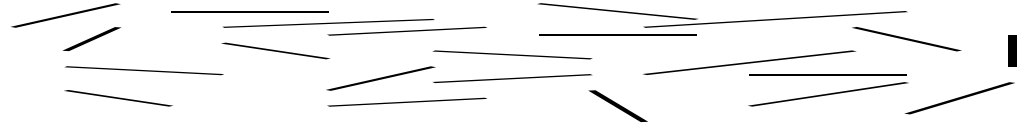
<https://github.com/schatzlab/appliedgenomics2026/tree/main/assignments/assignment2>
Due end of day on Monday Sept 28 (right before midnight)



0. Recap

Assembling a Genome

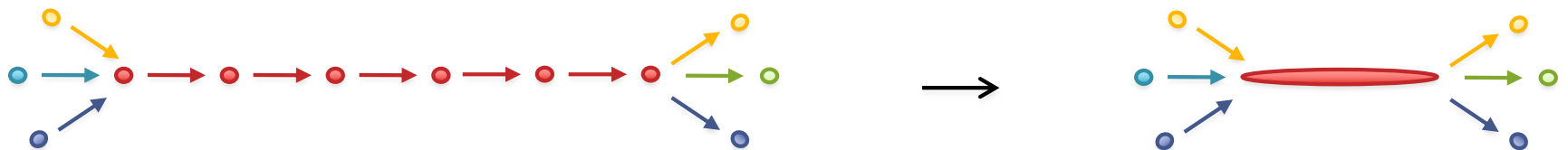
1. Shear & Sequence DNA



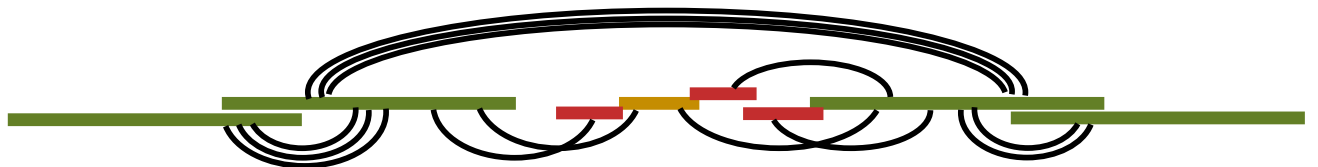
2. Construct assembly graph from reads (de Bruijn / overlap graph)

...AGCCTAGGGATGCGCGACACGT
GGATGCGCGACACGT CGCATATCCGGTTTGGT CAACCTCGGACGGAC
CAACCTCGGACGGACCTCAGCGAA...

3. Simplify assembly graph

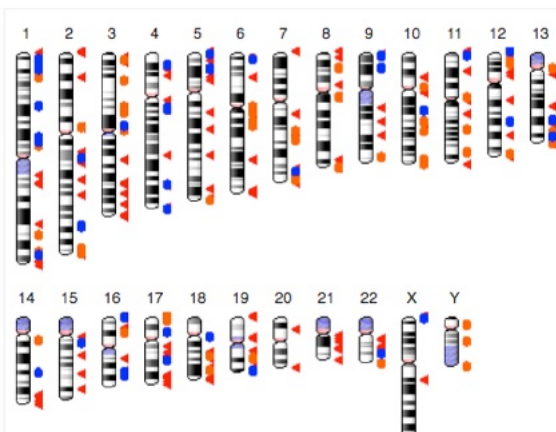


4. Detangle graph with long reads, mates, and other links



Human Genome Overview

Information about the continuing improvement of the human genome



- ◀ Region containing alternate loci
- ◀ Region containing fix patches
- ◀ Region containing novel patches

Ideogram of the latest human assembly, GRCh38.p11

The GRC is working hard to provide the best possible by both generating multiple representations (alternates) represented by a single path. Additionally, we are releasing allows users who are interested in a specific locus to affecting users who need chromosome coordinate s

Download data:

- [GRCh38.p11 \(latest minor release\) FTP](#)
- [GRCh38 \(latest major release\) FTP](#)
- [Genomic regions under review FTP](#)
- [Current Tiling Path Files \(TPFs\)](#)

Transitioning to GRCh38? Try the [NCBI Remapping](#) assembly alignments used by the GRC.

Next assembly update

The next assembly update (GRCh38.p12) will be


[GRCh38.p11](#)
[GRCh37.p13](#)
[GRCh37](#)

GRCh38.p11

Release date: June 14, 2017

Release type: minor

Release notes: GRCh38.p11 is the eleventh patch release for the GRCh38 reference assembly. No chromosome coordinate of patch scaffolds is now: 64 FIX and 59 NOVEL.

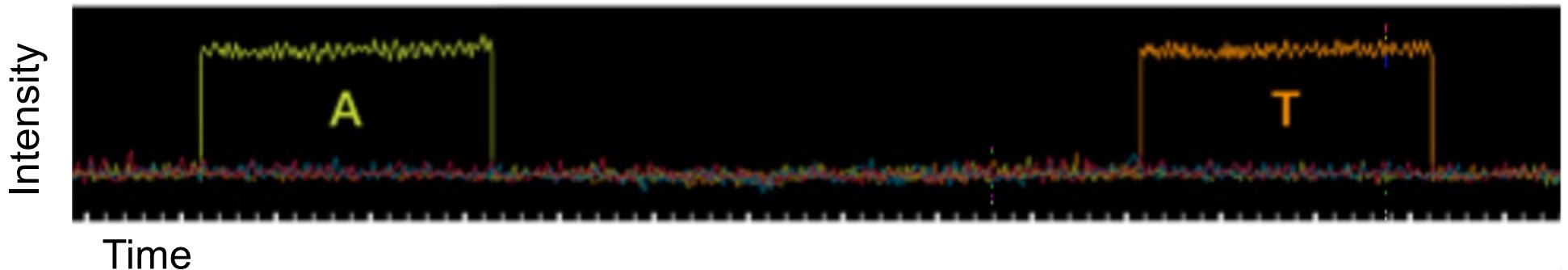
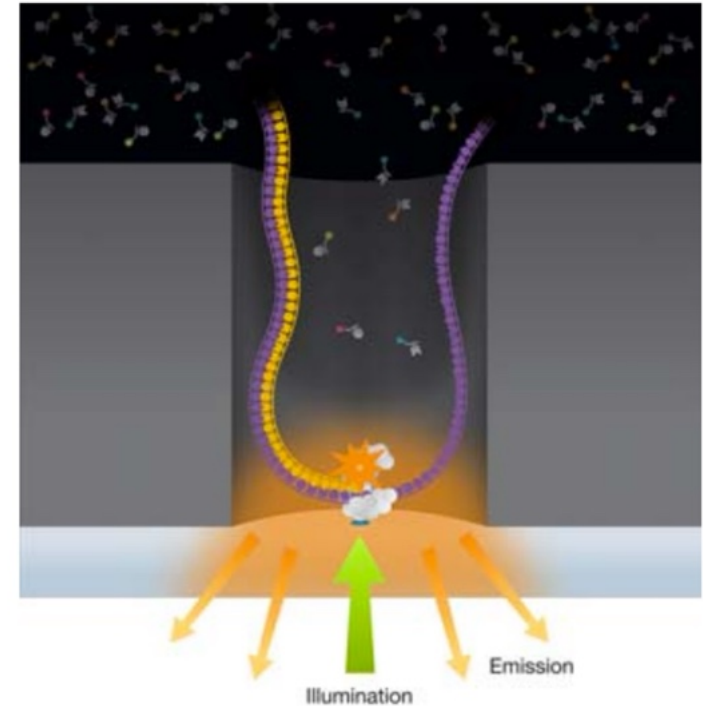
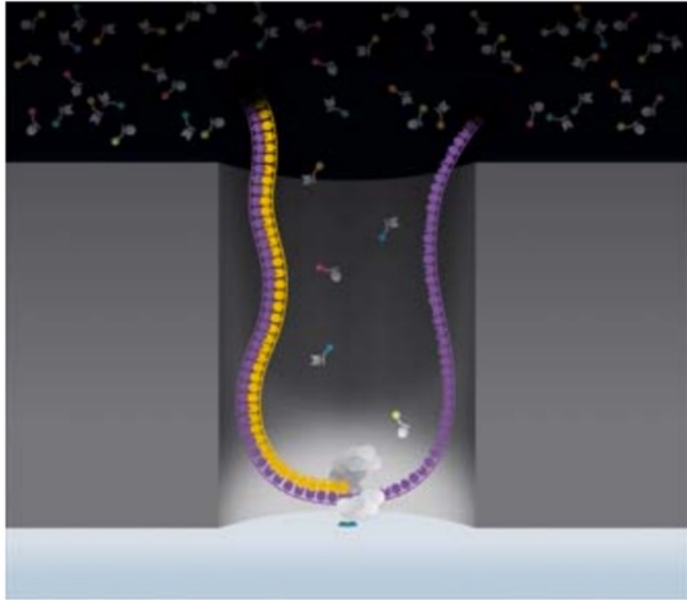
Assembly accessions: GenBank: [GCA_000001405.26](#), RefSeq: [GCF_000001405.37](#)

Pseudoautosomal regions

Name	Chr	Start	Stop
PAR#1	X	10,001	2,781,479
PAR#2	X	155,701,383	156,030,895
PAR#1	Y	10,001	2,781,479
PAR#2	Y	56,887,903	57,217,415

PacBio: SMRT Sequencing

Imaging of florescent phospholinked labeled nucleotides as they are incorporated by a polymerase anchored to a Zero-Mode Waveguide (ZMW).



“HiFi” Circular Consensus Reads

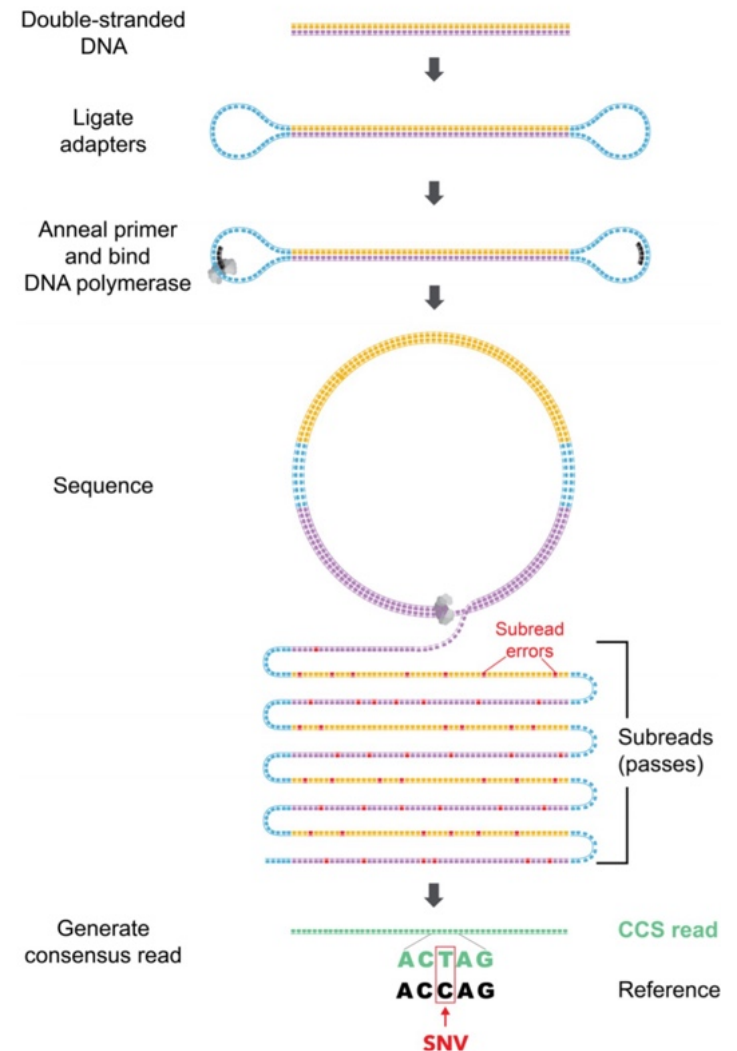
High-quality reads produced by sequencing the same molecule multiple times

Higher accuracy for low-coverage sequences like somatic variants or lowly expressed transcripts in RNA-seq, more interpretable alignments, better & faster assembly

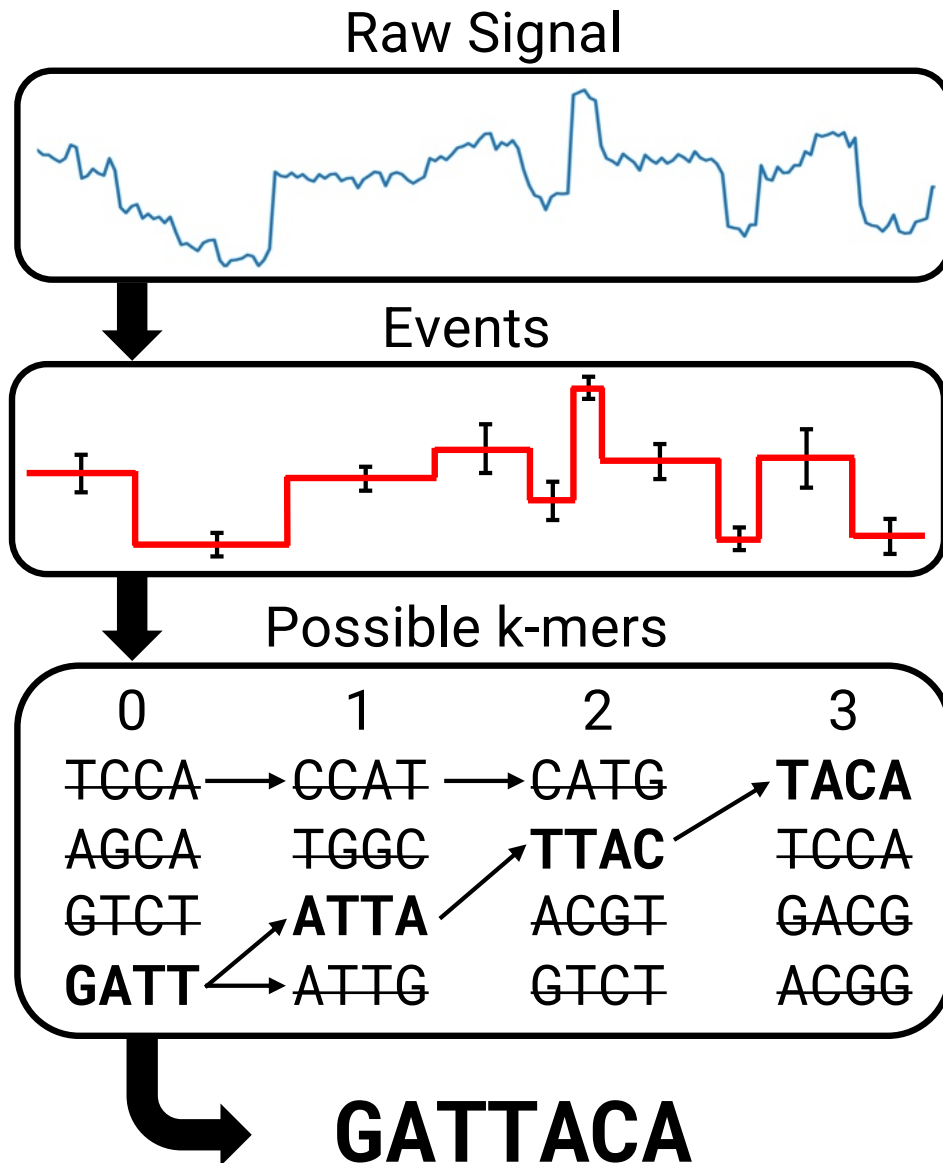
Limits read length, used to be very expensive but more manageable now

Accurate circular consensus long-read sequencing improves variant detection and assembly of a human genome

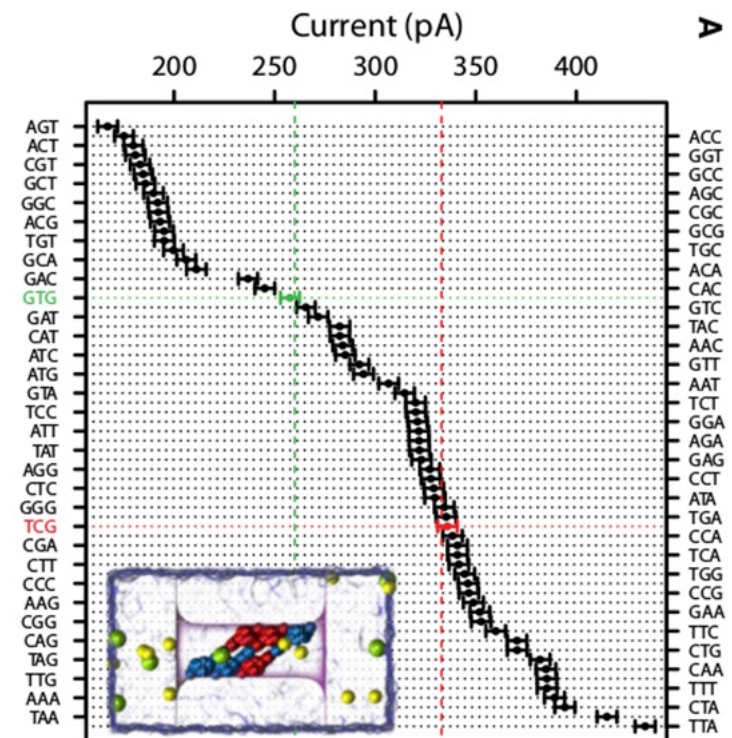
Wenger et al (2019) Nature Biotechnology doi:10.1038/s41587-019-0217-9



Nanopore Basecalling



Final sequence determined by most probable k-mers



"DNA Base-Calling from a Nanopore Using a Viterbi Algorithm"
Timp et al. (2012) *Biophysical Journal*

Let's finish a human genome



T2T Working Group

[Home](#)

[Technology](#)

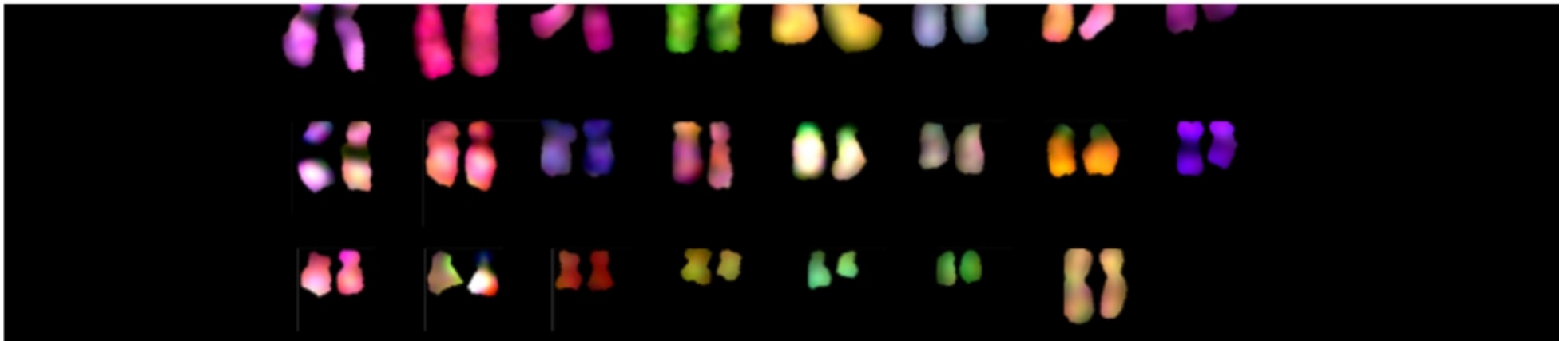
[Data](#)

[CHM13 Cell Line](#)

[Remaining Challenges](#) ▼

[Who We Are](#)

[Join Us](#)



The Telomere-to-Telomere (T2T) consortium is an open, community-based effort to generate the first complete assembly of a human genome.

CHM13 homozygous 46,XX cell line from Urvashi Surti, Pitt; SKY karyotype from Jennifer Gerton, Stowers



Science

HOME > COLLECTIONS > COMPLETING THE HUMAN GENOME

COMPLETING THE HUMAN GENOME

A fully sequenced human genome was announced more than 20 years ago. However, owing to technological limitations, some genomic regions remained unresolved. Here, *Science* and other journals present research by the Telomere-to-Telomere (T2T) Consortium, reporting on the endeavor to complete a comprehensive human reference genome.

6 RESULTS FOUND

SPECIAL ISSUE RESEARCH ARTICLE

Segmental duplications and their variation in a complete human genome

BY MITCHELL R. VOLLGER, XAVI GUITART, PHILIP C. DISHUCK, LUDOVICA MERCURI, WILLIAM T. HARVEY, ARIEL GERSHMAN, MARK DIEKHANS, ARVIS SULDOVARI, KATHERINE M. MUNSON, ALEXANDRA P. LEWIS, [...] EVAN E. EICHLE

SCIENCE • VOL. 376, NO. 6588 • 01 APR 2022

SPECIAL ISSUE RESEARCH ARTICLE

Complete genomic and epigenetic maps of human centromeres

BY NICOLAS ALTEMOSE, GLENNIS A. LOGSDON, ANDREY V. BZIKADZE, PRADYA SIKHWANI, SASHA A. LANGLEY, GINA Y. CALDAS, SAVANNAH J. HOYT, LEV URALSKY, FEDOR D. RYABOV, COLIN J. SHEW, [...] KAREN H. MIGA

SCIENCE • VOL. 376, NO. 6588 • 01 APR 2022

SPECIAL ISSUE RESEARCH ARTICLE

From telomere to telomere: The transcriptional and epigenetic state of human repeat elements

BY SAVANNAH J. HOYT, JESSICA M. STORER, GABRIELLE A. HARTLEY, PATRICK G. S. GRADY, ARIEL GERSHMAN, LEONARDO G. DE LIMA, CHARLES LIMOUSE, REZA HALABIAN, LUKE WOJENSKI, MATIAS RODRIGUEZ, [...] RACHEL J. C

+16 authors • SCIENCE • VOL. 376, NO. 6588 • 01 APR 2022

SPECIAL ISSUE RESEARCH ARTICLE

A complete reference genome improves analysis of human genetic variation

BY SERGEY AGANEZOV, STEPHANIE M. YAN, DANIELA C. SOTO, MELANIE KIRSCHKE, SAMANTHA ZARATE, PAVEL AVDEYEV, DYLAN J. TAYLOR, KISHWAR SHAFIK, ALAINE SHUMATE, CHUNLIN XIAO, [...] MICHAEL C. SCHATZ

SCIENCE • VOL. 376, NO. 6588 • 01 APR 2022

SPECIAL ISSUE RESEARCH ARTICLE

Epigenetic patterns in a complete human genome

BY ARIEL GERSHMAN, MICHAEL E. G. SAURIA, XAVI GUITART, MITCHELL R. VOLLGER, PAUL W. HOOK, SAVANNAH J. HOYT, MITEN JAIN, ALAINE SHUMATE, ROHAM RAZAGHI, SERGEY KOREN, [...] WINSTON TIMP

VOL. 376, NO. 6588 • 01 APR 2022

SPECIAL ISSUE RESEARCH ARTICLE

The complete sequence of a human genome

BY SERGEY NURK, SERGEY KOREN, ARANG RHIE, MIKKO RAUTIAINEN, ANDREY V. BZIKADZE, ALLA MIKHEENKO, MITCHELL R. VOLLGER, NICOLAS ALTEMOSE, LEV URALSKY, ARIEL GERSHMAN, [...] ADAM M. PHILLIPPY

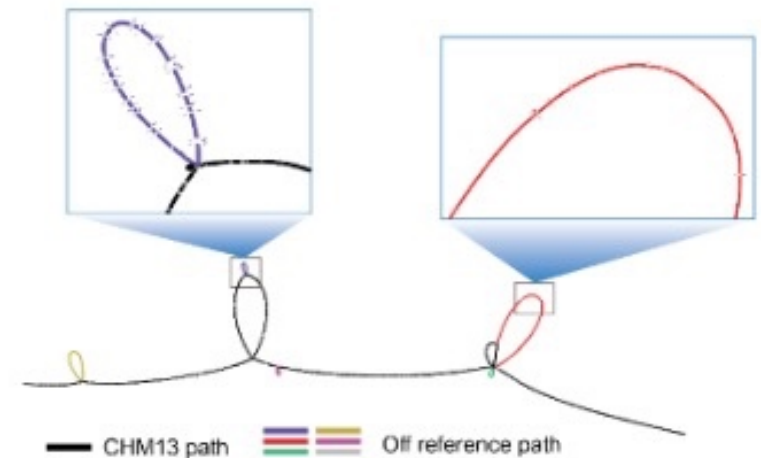
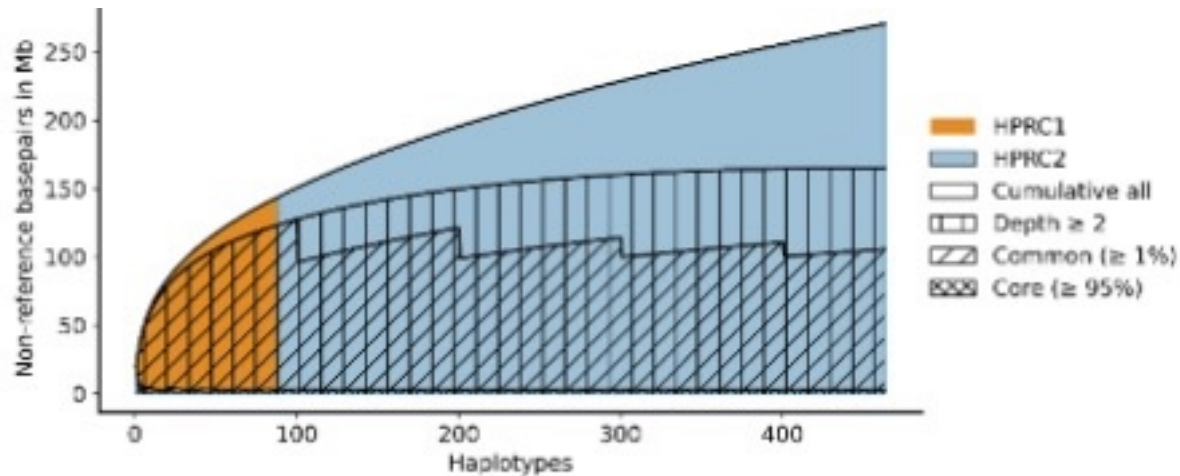
SCIENCE • VOL. 376, NO. 6588 • 31 MAR 2022 • 44-53



Human Pangenome Reference Consortium (HPRC)

HPRC2 contributes 460 haplotypes that together capture over 99% of common variation

HPRC2 pangenome captures 2.6 million SNPs, 2.3 million indels and 80 thousand SVs that are off reference



HPRC2: A human pangenome reference with near-complete coverage of common genetic variation

Lucas *et al.* (2026) *bioRxiv*. <https://doi.org/10.64898/2026.07.21.739710>



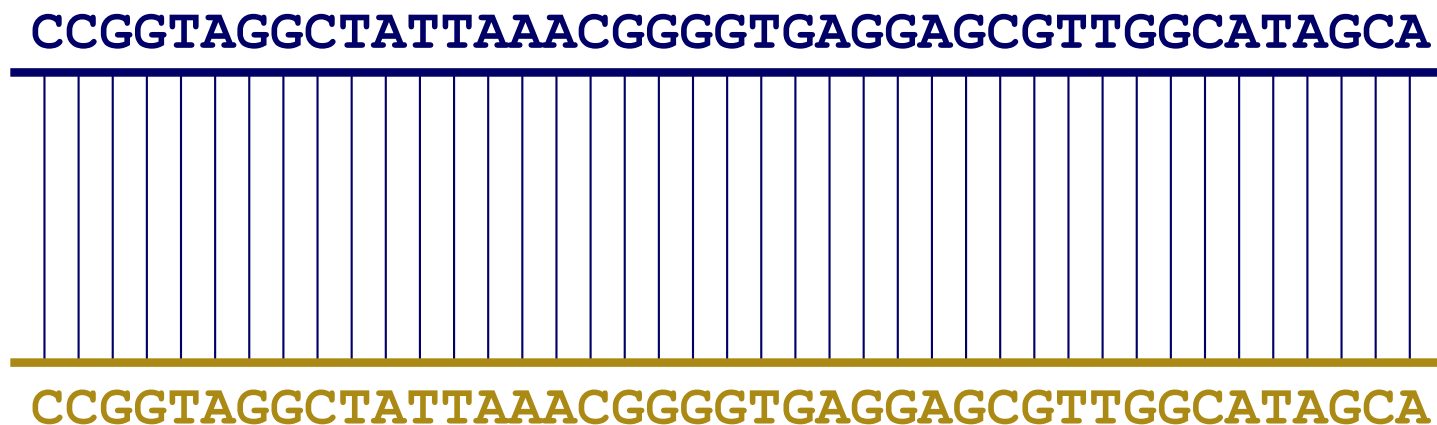
Part I.

Whole Genome Alignment

Slides Courtesy of Adam M. Phillippy
NHGRI & JHU

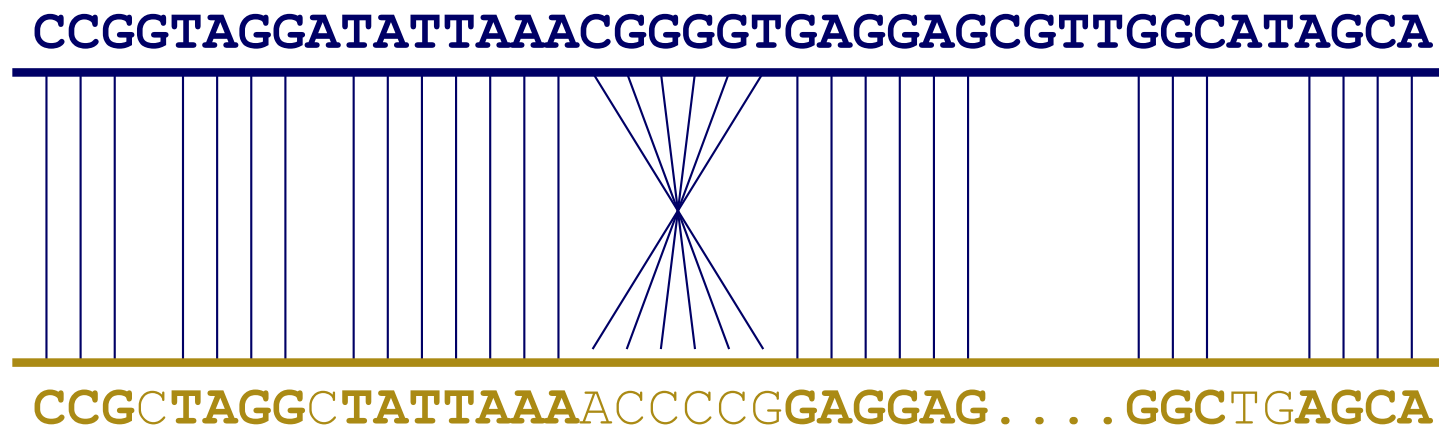
Goal of WGA

- For two genomes, A and B , find a mapping from each position in A to its corresponding position in B



Not so fast...

- Genome *A* may have insertions, deletions, translocations, inversions, duplications or SNPs with respect to *B* (sometimes all of the above)



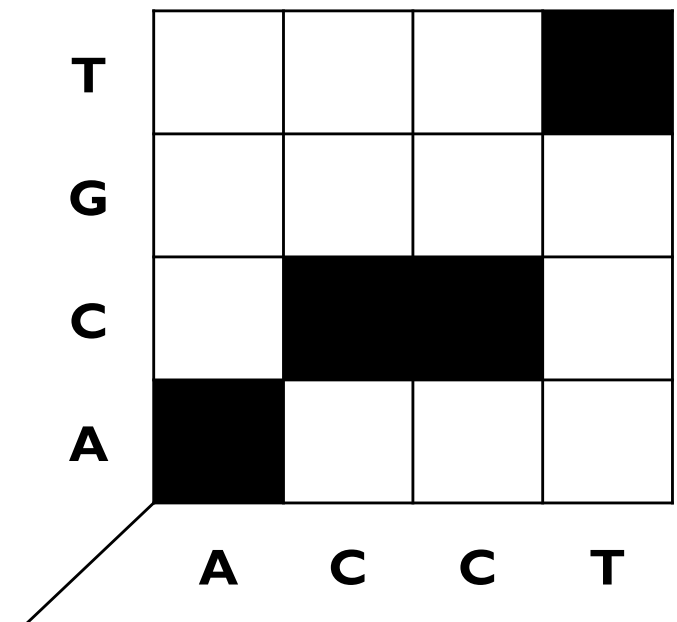
WGA visualization

- How can we visualize *whole* genome alignments?

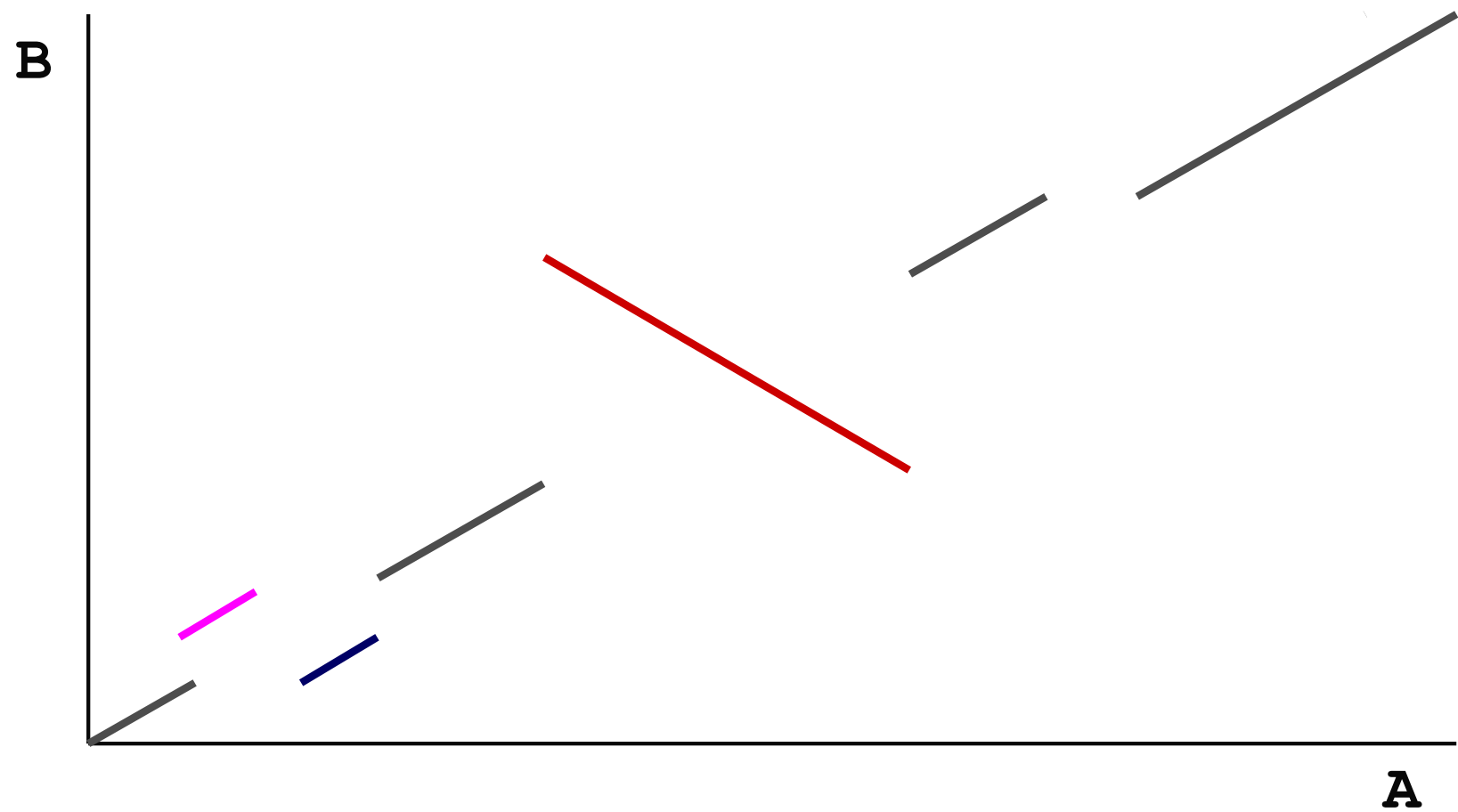
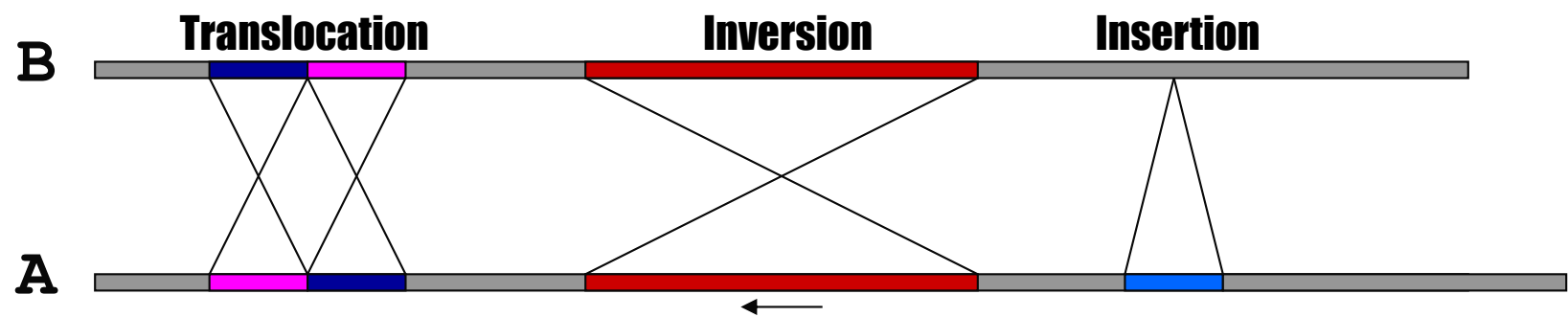
- With an alignment dot plot

- $N \times M$ matrix

- Let i = position in genome A
 - Let j = position in genome B
 - Fill cell (i,j) if A_i shows similarity to B_j



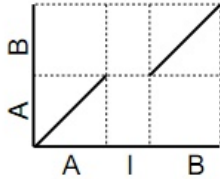
- A perfect alignment between A and B would completely fill the positive diagonal



SV Types

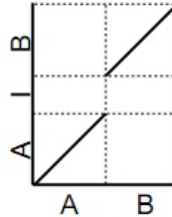
Insertion into Reference

R: AIB
Q: AB



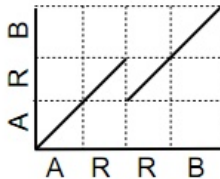
Insertion into Query

R: AB
Q: AIB



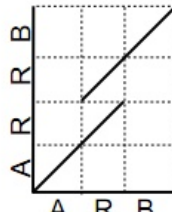
Collapse Query

R: ARRB
Q: ARB



Collapse Reference

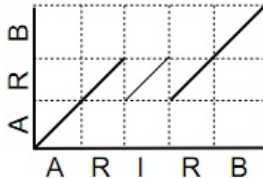
R: ARB
Q: ARRB



Collapse Query
w/ Insertion

R: ARIRB
Q: ARB

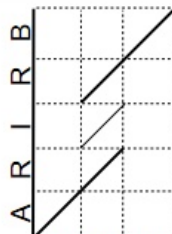
Exact tandem
alignment if I=R



Collapse Reference
w/ Insertion

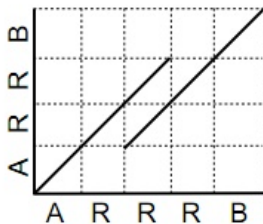
R: ARB
Q: ARIRB

Exact tandem
alignment if I=R



Collapse Query

R: ARRRB
Q: ARRB



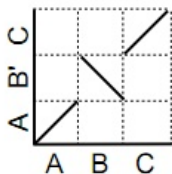
Collapse Reference

R: ARRB
Q: ARRRB



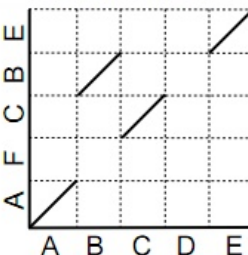
Inversion

R: ABC
Q: AB'C



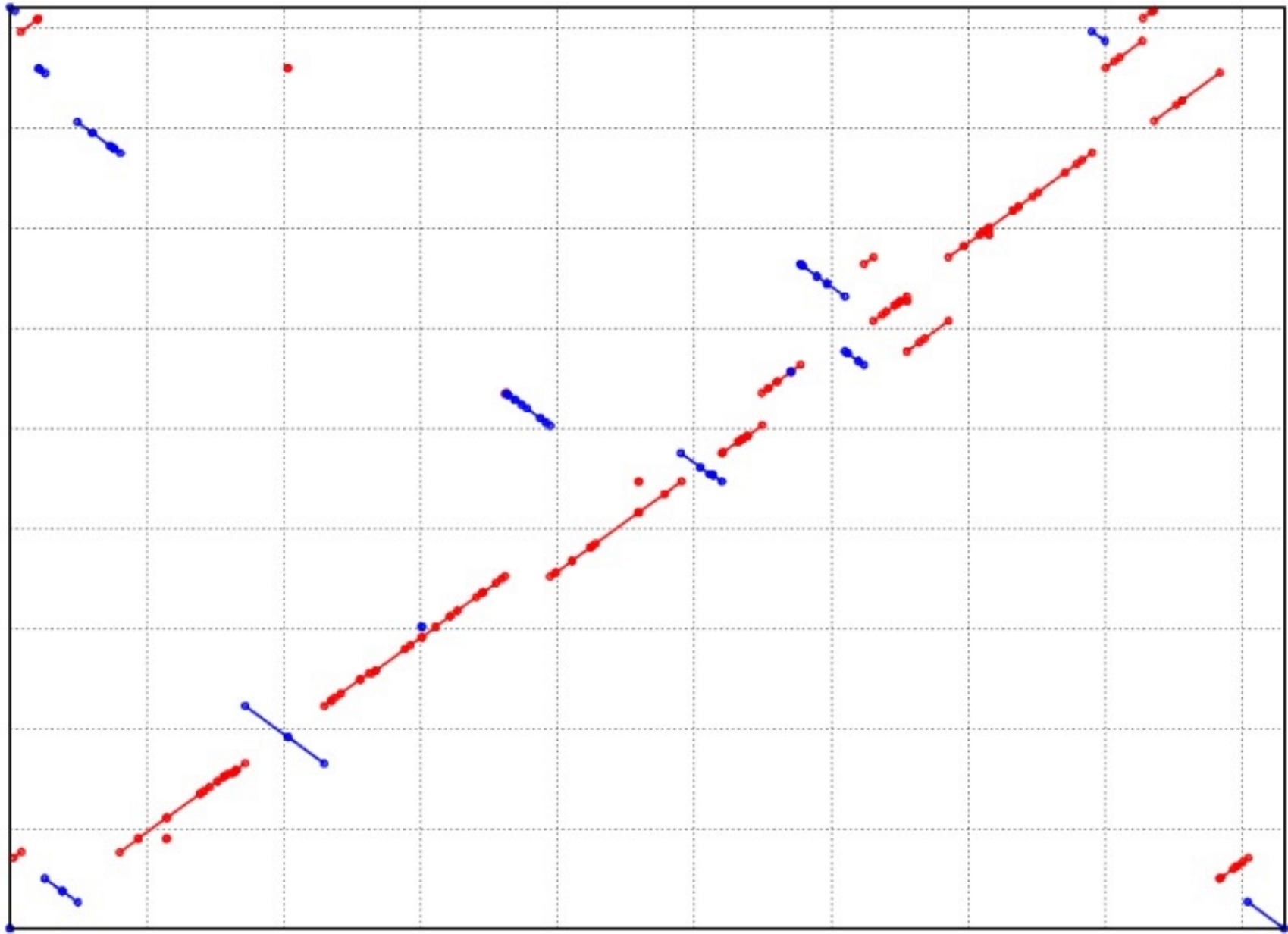
Rearrangement
w/ Disagreement

R: ABCDE
Q: AFCBE



- Different structural variation types / misassemblies will be apparent by their pattern of breakpoints
- Most breakpoints will be at or near repeats
- Things quickly get complicated in real genomes

<http://mummer.sf.net/manual/AlignmentTypes.pdf>



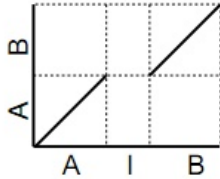
Alignment of 2 strains of *Y. pestis*

<http://mummer.sourceforge.net/manual/>

SV Types

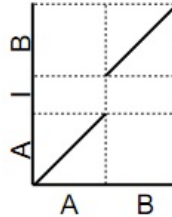
Insertion into Reference

R: AIB
Q: AB



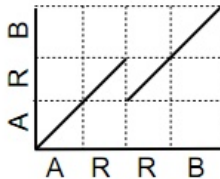
Insertion into Query

R: AB
Q: AIB



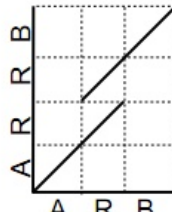
Collapse Query

R: ARRB
Q: ARB



Collapse Reference

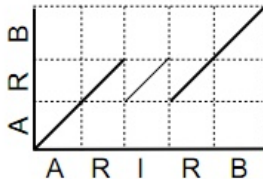
R: ARB
Q: ARRB



Collapse Query
w/ Insertion

R: ARIRB
Q: ARB

Exact tandem
alignment if I=R



Collapse Reference
w/ Insertion

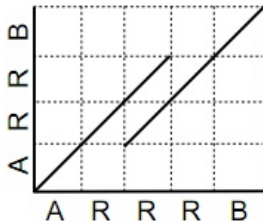
R: ARB
Q: ARIRB

Exact tandem
alignment if I=R



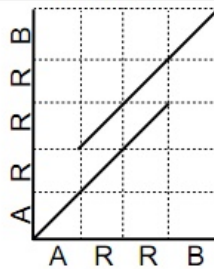
Collapse Query

R: ARRRB
Q: ARRB



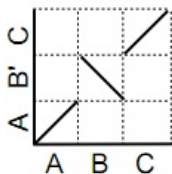
Collapse Reference

R: ARRB
Q: ARRRB



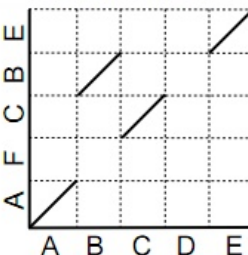
Inversion

R: ABC
Q: AB'C



Rearrangement
w/ Disagreement

R: ABCDE
Q: AFCBE



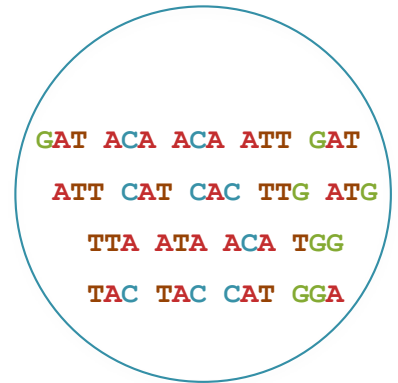
- Different structural variation types / misassemblies will be apparent by their pattern of breakpoints
- Most breakpoints will be at or near repeats
- Things quickly get complicated in real genomes

<http://mummer.sf.net/manual/AlignmentTypes.pdf>

GATTACATACACATTGGATG

GATTACATACACATTGGATG

GATTACATACACATTGGATG
 GAT ACA ACA ATT GAT
 ATT CAT CAC TTG ATG
 TTA ATA ACA TGG
 TAC TAC CAT GGA

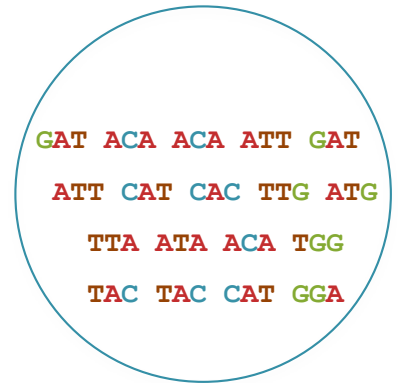


=

GATTACATACACATTGGATG
 GAT ACA ACA ATT GAT
 ATT CAT CAC TTG ATG
 TTA ATA ACA TGG
 TAC TAC CAT GGA



GATTACATACACATTGGATG
 GAT ACA ACA ATT GAT
 ATT CAT CAC TTG ATG
 TTA ATA ACA TGG
 TAC TAC CAT GGA



=

GATTACATACACATTGGATG
 GAT ACA ACA ATT GAT
 ATT CAT CAC TTG ATG
 TTA ATA ACA TGG
 TAC TAC CAT GGA



Jaccard Coefficient

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{18}{18} = 100\%$$

GATTACATACACATTGGATG

GATTACATACXCATTTGGATG

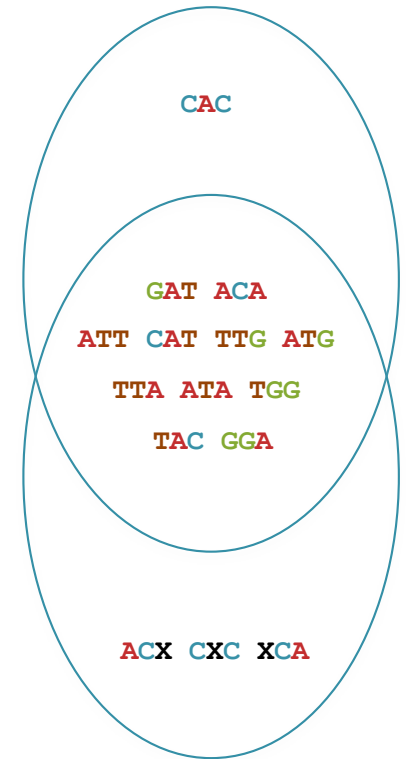
G	A	T	A	C	A	T	A	C	A	C	A	T	T	G	G	A	T	G	
G	A	T		A	C	A		A	C	A		A	T	T		G	A	T	
	A	T	T		C	A	T		C	A	C		T	T	G		A	T	G
		T	T	A		A	T	A		A	C	A		T	G	G			
			T	A	C		T	A	C		C	A	T			G	G	A	

G	A	T	A	C	A	T	A	C	X	C	A	T	T	G	G	A	T	G	
G	A	T		A	C	A		A	C	X		A	T	T		G	A	T	
	A	T	T		C	A	T		C	X	C		T	T	G		A	T	G
		T	T	A		A	T	A		X	C	A		T	G	G			
			T	A	C		T	A	C		C	A	T			G	G	A	



GATTACATACACAATTGGATG
 GAT ACA ACA ATT GAT
 ATT CAT CAC TTG ATG
 TTA ATA ACA TGG
 TAC TAC CAT GGA

GATTACATACXCAATTGGATG
 GAT ACA ACX ATT GAT
 ATT CAT CX C TTG ATG
 TTA ATA XCA TGG
 TAC TAC CAT GGA

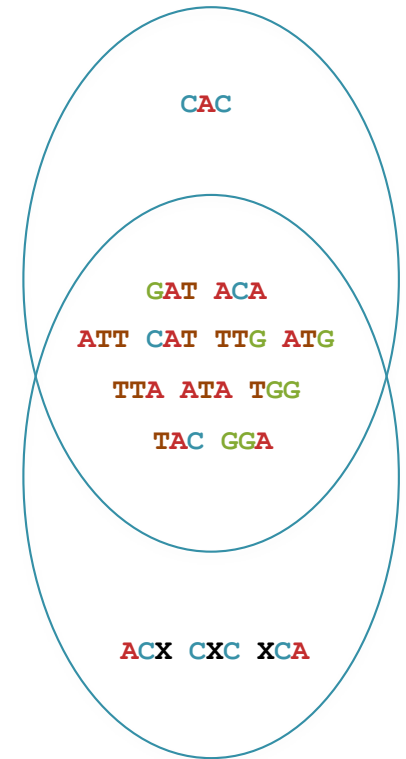


Jaccard Coefficient

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{11}{15} = 73.3\%$$

GATTACATACACAATTGGATG
 GAT ACA ACA ATT GAT
 ATT CAT CAC TTG ATG
 TTA ATA ACA TGG
 TAC TAC CAT GGA

GATTACATACXCAATTGGATG
 GAT ACA ACX ATT GAT
 ATT CAT CX C TTG ATG
 TTA ATA XCA TGG
 TAC TAC CAT GGA



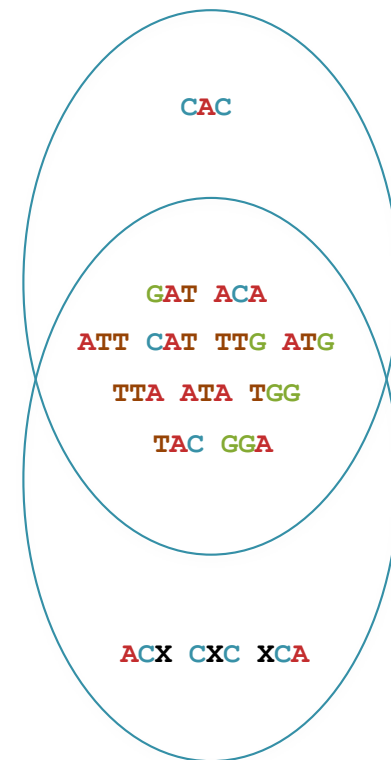
$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

$|A| = |B| = N$
 $|A \cap B| = pN$
 $|A \cup B| = |A| + |B| - |A \cap B| = N + N - pN = (2 - p)N$

p = prob(kmer shared)

GATTACATACACAATTGGATG
 GAT ACA ACA ATT GAT
 ATT CAT CAC TTG ATG
 TTA ATA ACA TGG
 TAC TAC CAT GGA

GATTACATACXCAATTGGATG
 GAT ACA ACX ATT GAT
 ATT CAT CX C TTG ATG
 TTA ATA XCA TGG
 TAC TAC CAT GGA



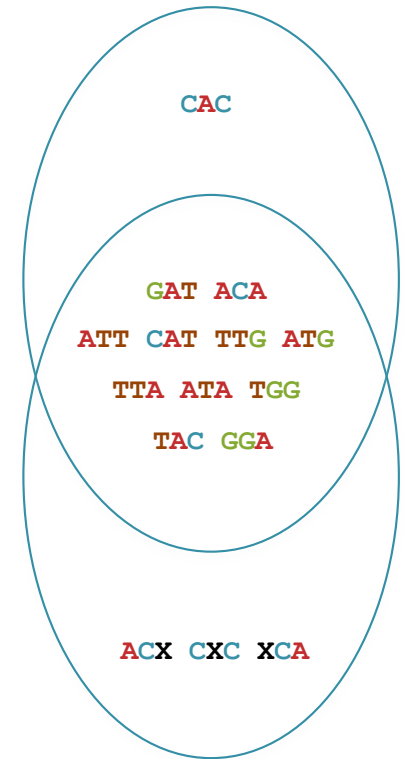
$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

$$J \approx \frac{p}{2 - p}$$

p = prob(kmer shared)

GATTACATACACAATTGGATG
 GAT ACA ACA ATT GAT
 ATT CAT CAC TTG ATG
 TTA ATA ACA TGG
 TAC TAC CAT GGA

GATTACATACXCAATTGGATG
 GAT ACA ACX ATT GAT
 ATT CAT CX C TTG ATG
 TTA ATA XCA TGG
 TAC TAC CAT GGA

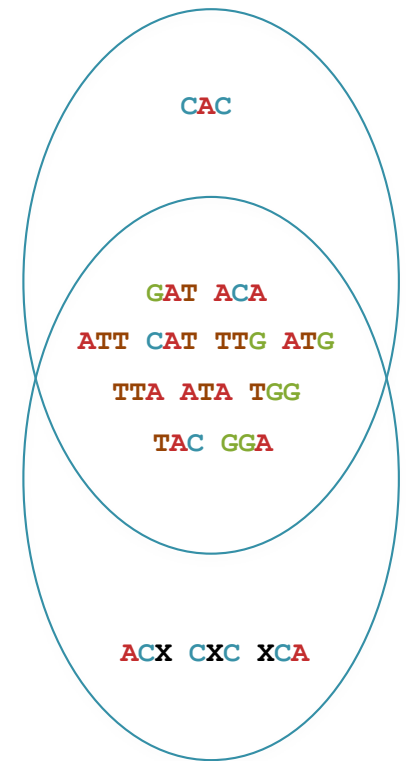


From Jaccard to Average Nucleotide Identity

$$J \approx \frac{p}{2 - p} \implies p = \frac{2J}{1 + J}$$

GATTACATACACAATTGGATG
 GAT ACA ACA ATT GAT
 ATT CAT CAC TTG ATG
 TTA ATA ACA TGG
 TAC TAC CAT GGA

GATTACATACXCAATTGGATG
 GAT ACA ACX ATT GAT
 ATT CAT CX C TTG ATG
 TTA ATA XCA TGG
 TAC TAC CAT GGA



From Jaccard to Average Nucleotide Identity

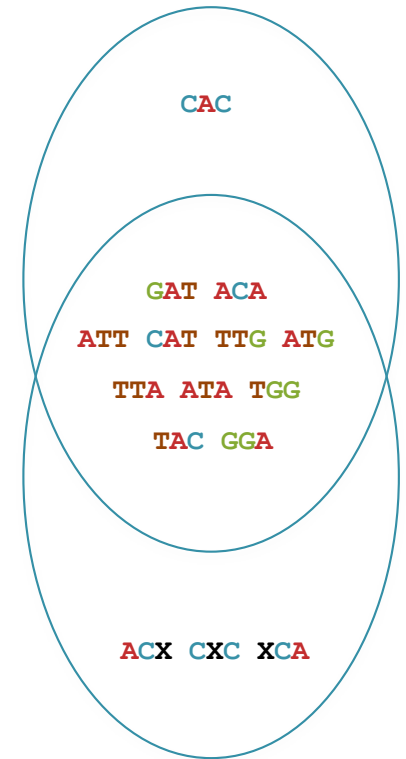
$$J \approx \frac{p}{2-p} \implies p = \frac{2J}{1+J}$$

$$p = I^k$$

$$I = \text{ANI} = p^{1/k} = \left(\frac{2J}{1+J} \right)^{1/k}$$

GATTACATACACAATTGGATG
 GAT ACA ACA ATT GAT
 ATT CAT CAC TTG ATG
 TTA ATA ACA TGG
 TAC TAC CAT GGA

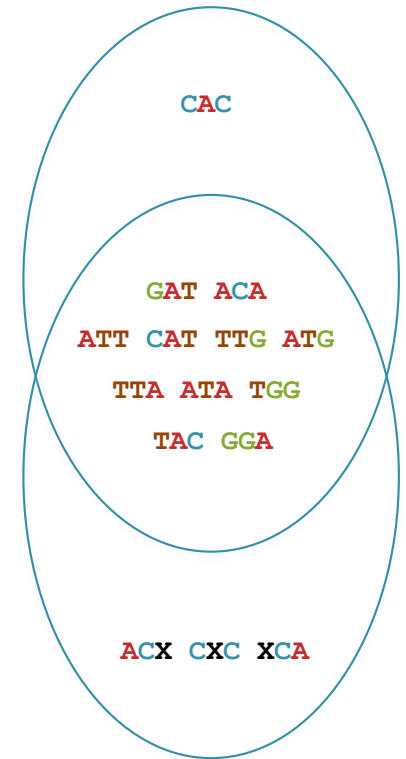
GATTACATACXCAATTGGATG
 GAT ACA ACX ATT GAT
 ATT CAT CX C TTG ATG
 TTA ATA XCA TGG
 TAC TAC CAT GGA



Identity for small differences (d): $p = I^k = (1 - d)^k \approx e^{-kd}$

GATTACATACACAATTGGATG
 GAT ACA ACA ATT GAT
 ATT CAT CAC TTG ATG
 TTA ATA ACA TGG
 TAC TAC CAT GGA

GATTACATACXCAATTGGATG
 GAT ACA ACX ATT GAT
 ATT CAT CX C TTG ATG
 TTA ATA XCA TGG
 TAC TAC CAT GGA



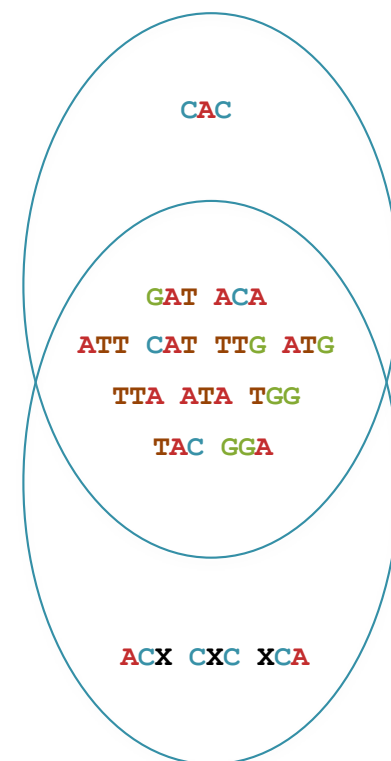
Identity for small differences (d): $p = I^k = (1 - d)^k \approx e^{-kd}$

$$p \approx e^{-kd} \Rightarrow d \approx -\frac{1}{k} \ln p \Rightarrow \text{ANI} = 1 - d \approx 1 + \frac{1}{k} \ln p.$$

$$\text{ANI} \approx 1 + \frac{1}{k} \ln \left(\frac{2J}{1 + J} \right)$$

GATTACATACACAATTGGATG
 GAT ACA ACA ATT GAT
 ATT CAT CAC TTG ATG
 TTA ATA ACA TGG
 TAC TAC CAT GGA

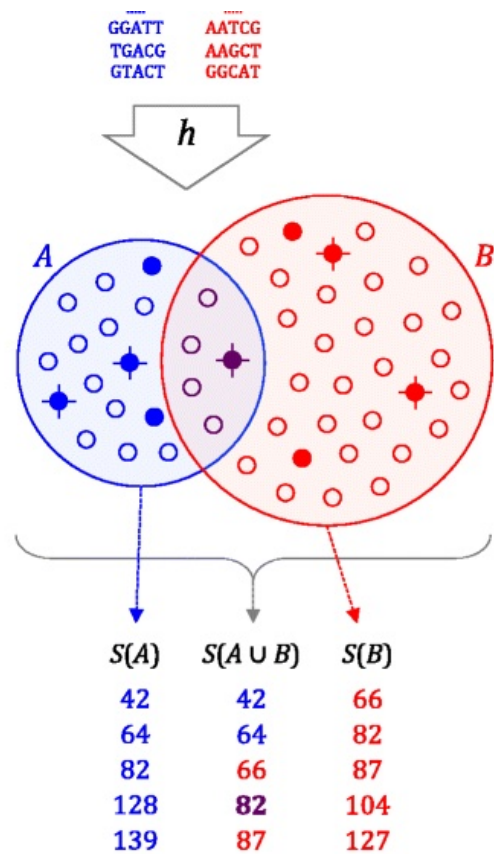
GATTACATACXCAATTGGATG
 GAT ACA ACX ATT GAT
 ATT CAT CX C TTG ATG
 TTA ATA XCA TGG
 TAC TAC CAT GGA



$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{11}{15} = 73.3\%$$

$$\text{ANI} \approx 1 + \frac{1}{k} \ln \left(\frac{2J}{1+J} \right)$$

ANI = 94.4%



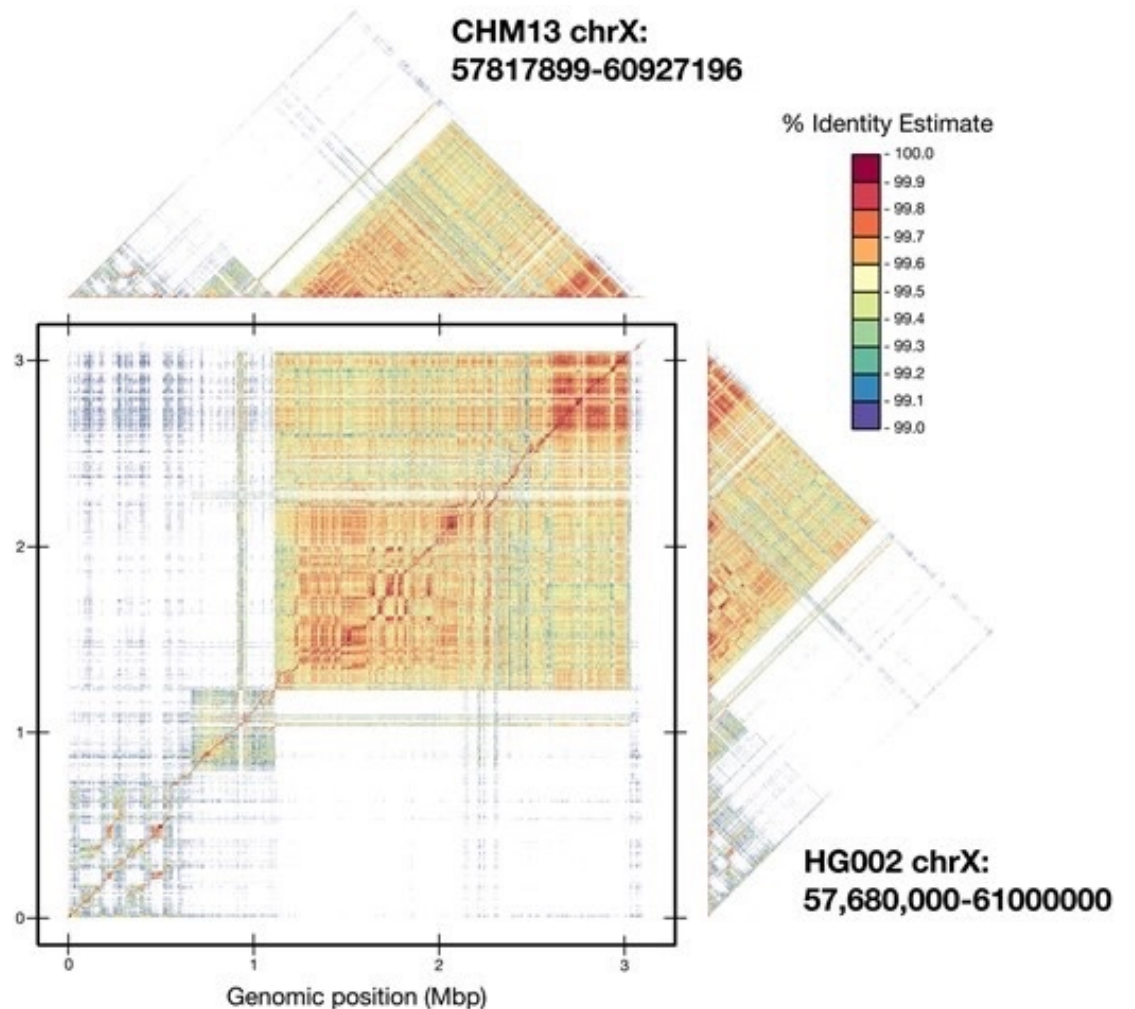
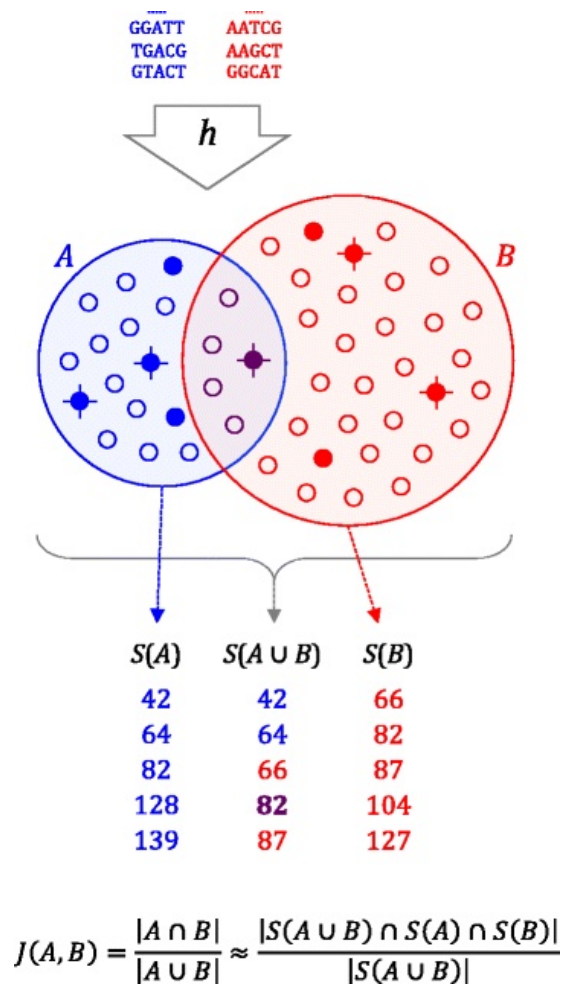
$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \approx \frac{|S(A \cup B) \cap S(A) \cap S(B)|}{|S(A \cup B)|}$$

Mash: fast genome and metagenome distance estimation using MinHash

Ondov et al (2016) Genome Biology. <https://doi.org/10.1186/s13059-016-0997-x>

ModDotPlot—rapid and interactive visualization of tandem repeats

Sweeten, Schatz, Phillippy (2024) Bioinformatics. <https://doi.org/10.1093/bioinformatics/btae493>

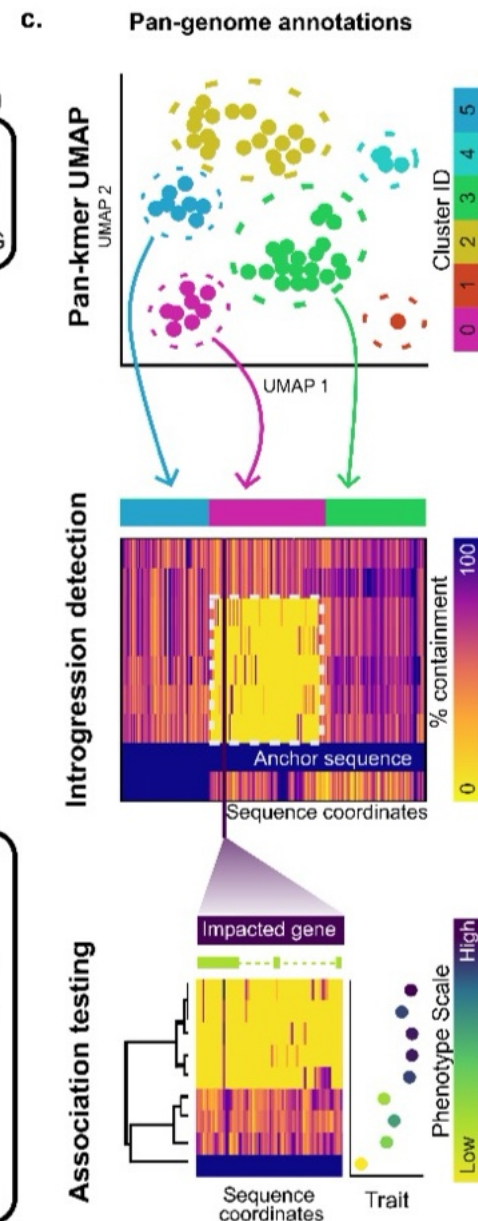
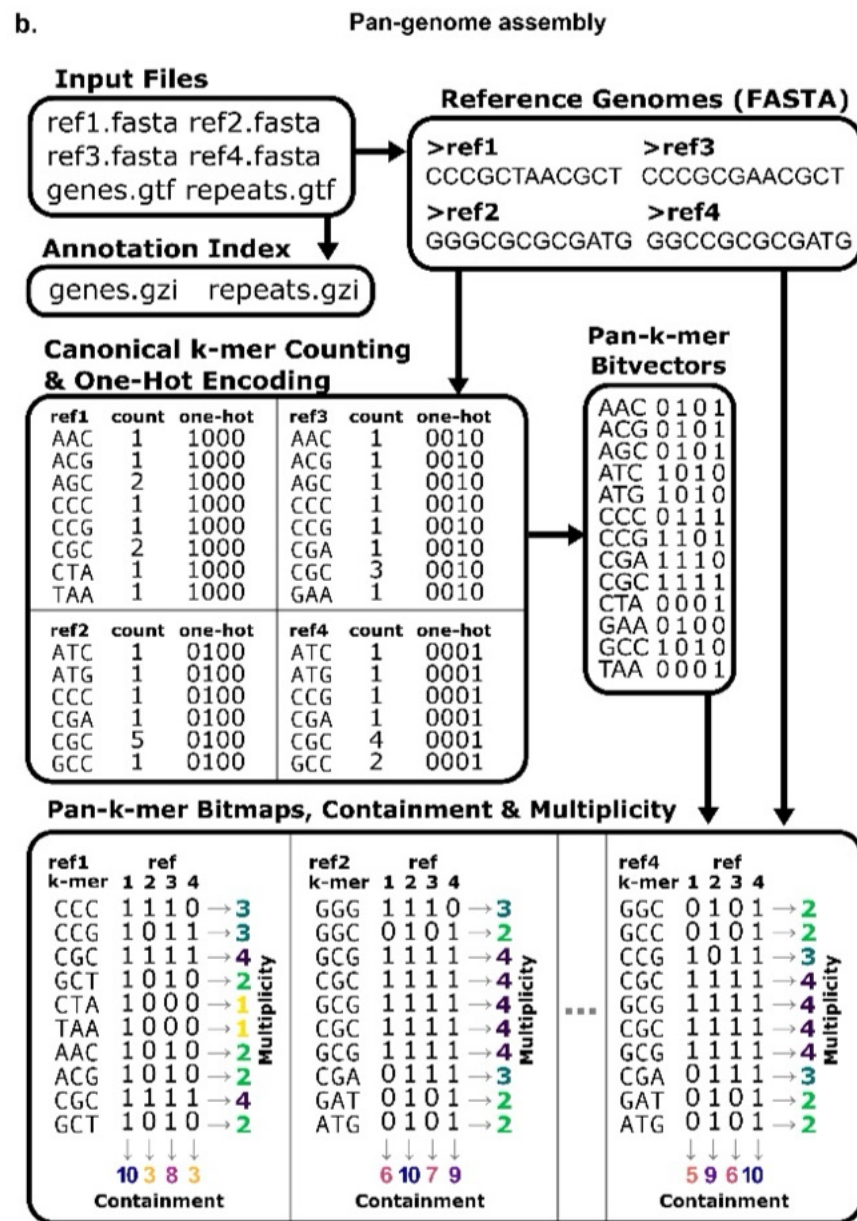
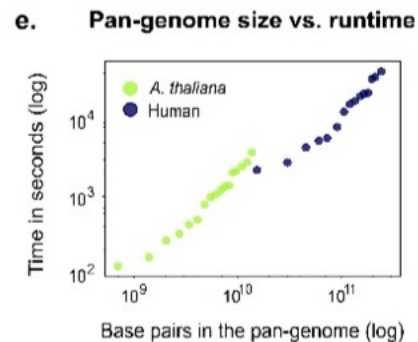
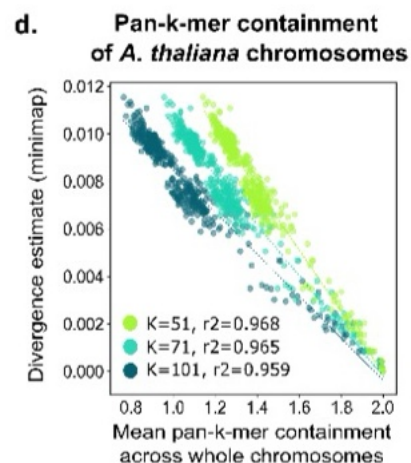
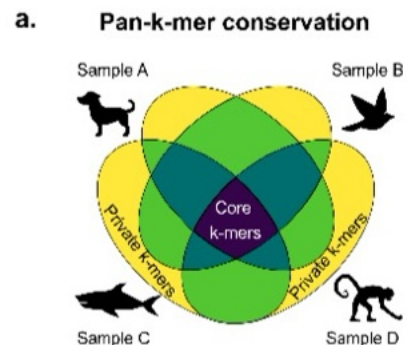


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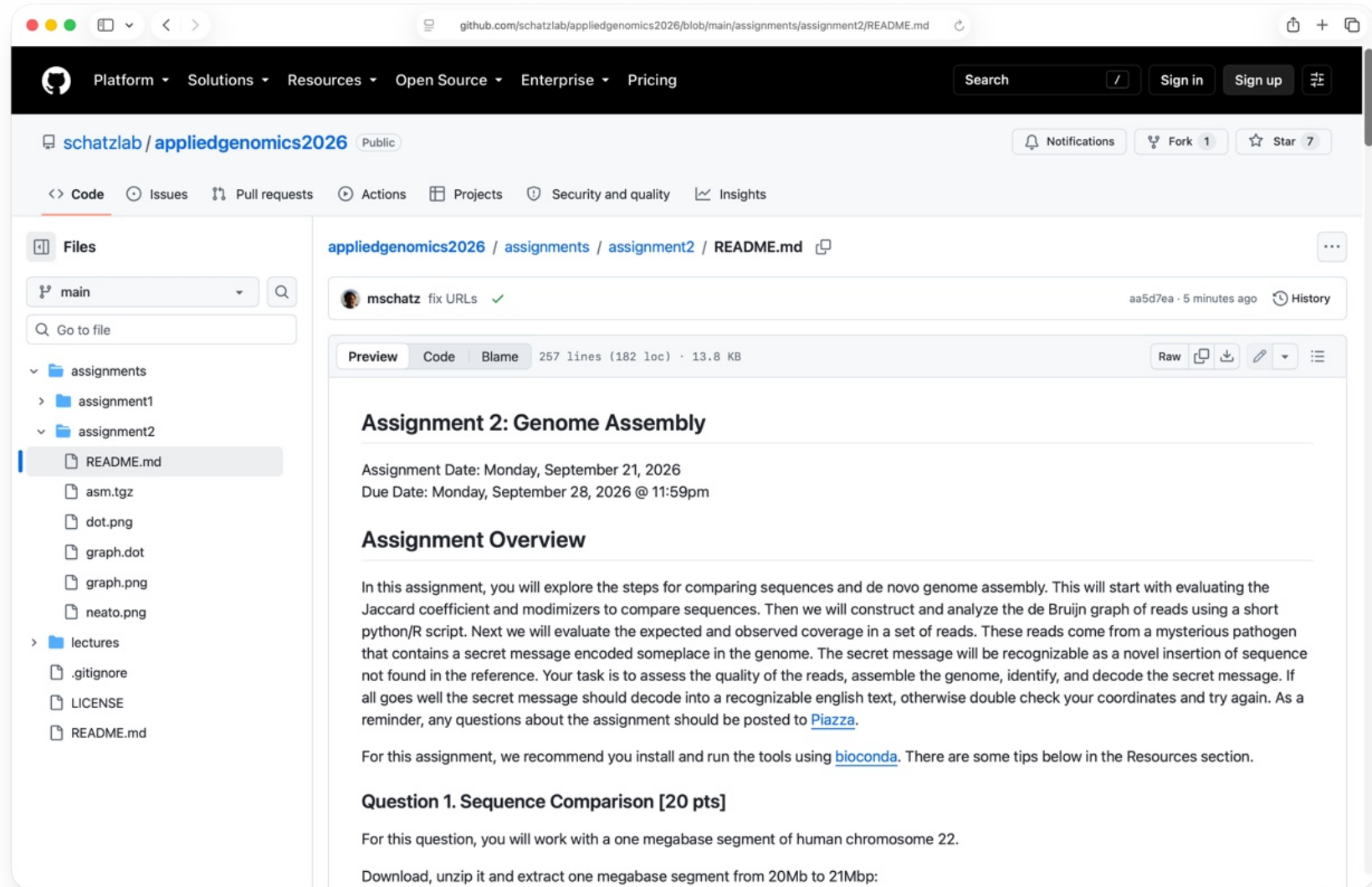
ModDotPlot—rapid and interactive visualization of tandem repeats

Sweeten, Schatz, Phillippy (2024) Bioinformatics. <https://doi.org/10.1093/bioinformatics/btae493>



Panagram reveals pangenome architectures underlying trait variation in plants and animals
Jenike, Brown, et al (2026) In submission!

Assignment 2



The screenshot shows a GitHub repository page for 'appliedgenomics2026'. The left sidebar displays a file tree with folders 'assignments', 'lectures', and files like 'README.md', 'asm.tgz', 'dot.png', 'graph.dot', 'graph.png', 'neato.png', '.gitignore', 'LICENSE', and 'README.md'. The main content area shows the 'assignment2/README.md' file, which includes the title 'Assignment 2: Genome Assembly', the assignment date (Monday, September 21, 2026), the due date (Monday, September 28, 2026 @ 11:59pm), and an 'Assignment Overview' section. The overview describes the task of comparing sequences and de novo genome assembly, mentioning the Jaccard coefficient, modimizers, and a secret message encoded in the genome. It also provides a recommendation to install and run tools using 'bioconda'.

Assignment 2: Genome Assembly

Assignment Date: Monday, September 21, 2026
Due Date: Monday, September 28, 2026 @ 11:59pm

Assignment Overview

In this assignment, you will explore the steps for comparing sequences and de novo genome assembly. This will start with evaluating the Jaccard coefficient and modimizers to compare sequences. Then we will construct and analyze the de Bruijn graph of reads using a short python/R script. Next we will evaluate the expected and observed coverage in a set of reads. These reads come from a mysterious pathogen that contains a secret message encoded someplace in the genome. The secret message will be recognizable as a novel insertion of sequence not found in the reference. Your task is to assess the quality of the reads, assemble the genome, identify, and decode the secret message. If all goes well the secret message should decode into a recognizable english text, otherwise double check your coordinates and try again. As a reminder, any questions about the assignment should be posted to [Piazza](#).

For this assignment, we recommend you install and run the tools using [bioconda](#). There are some tips below in the Resources section.

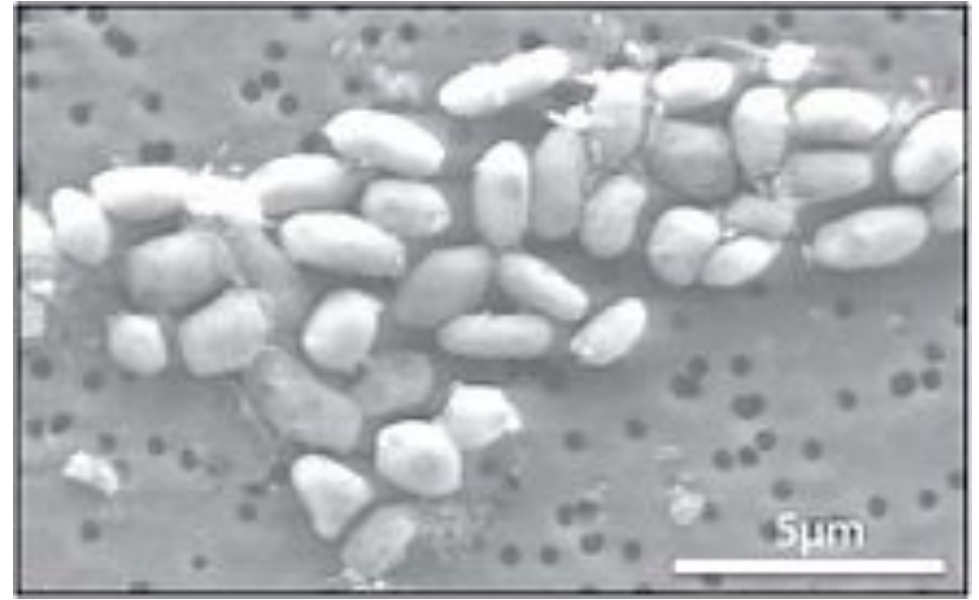
Question 1. Sequence Comparison [20 pts]

For this question, you will work with a one megabase segment of human chromosome 22.

Download, unzip it and extract one megabase segment from 20Mb to 21Mbp:

<https://github.com/schatzlab/appliedgenomics2026/tree/main/assignments/assignment2>
Due end of day on Monday Sept 28 (right before midnight)

Halomonas sp. GFAJ-1



Library 1: Fragment

Avg Read length: 100bp

Insert length: 180bp

Library 2: Short jump

Avg Read length: 50bp

Insert length: 2000bp

A Bacterium That Can Grow by Using Arsenic Instead of Phosphorus

Wolfe-Simon et al (2010) *Science*. 332(6034)1163-1166.

Digital Information Storage

Decoding self-referential DNA that encodes these notes.

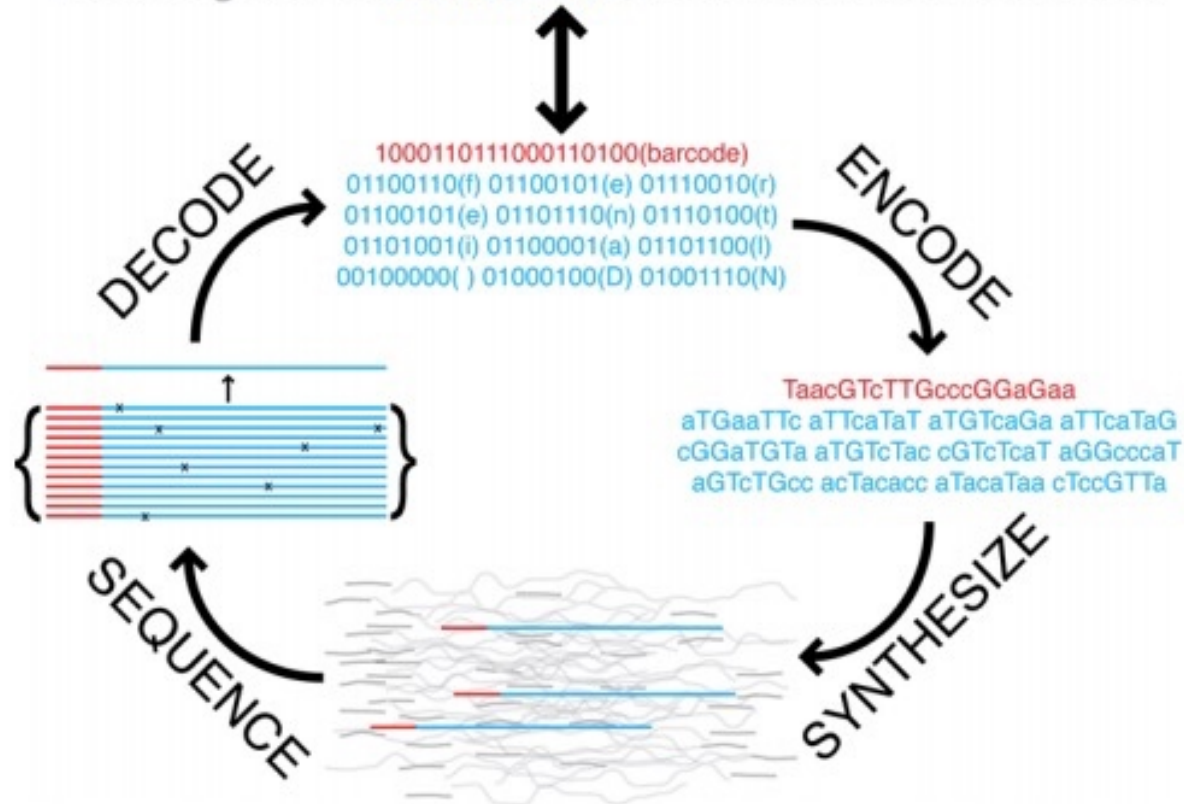


Fig. S1. Schematic of DNA information storage.

Encoding/decoding algorithm implemented in dna-encode.pl from David Dooling.

Next-generation Digital Information Storage in DNA

Church et al (2010) *Science*. 337(6102)1628

Assignment 2: Genome Assembly

Due Monday Sept 28 @ 11:59pm

- 1. Setup Conda/Docker/Ubuntu**
- 2. Initialize Tools**
- 3. Download Reference Genome & Reads**
- 4. Decode the secret message**
 1. Check kmer distribution
 2. Assemble the reads with spades
 3. Align to reference with MUMmer
 4. Extract foreign sequence
 5. dna-decode.py

<https://github.com/schatzlab/appliedgenomics2026/blob/master/assignments/assignment2/README.md>

